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Validation of a Mathematical Model of a Hydrogen Powered Scale Model Train

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Abstract

The decarbonisation of the railway sector has increased interest in alternative traction systems, including hydrogen fuel-cell powertrains for routes where conventional electrification is not feasible. Before such systems can be used to investigate future hybrid architectures and Energy Management Strategies, their simulation models must represent the interaction between mechanical loading and electrical source behaviour. This project addresses that need through the physical validation of a mathematical model for a hydrogen-powered G-scale model train.

An existing baseline train model was converted into a coupled electromechanical simulation in which the train dynamics, traction motors, drivetrain, mechanical resistance, and load-dependent fuel-cell behaviour operate as an integrated system. The purpose of this migration was to replace the idealised power-source behaviour of the baseline model with a representation capable of responding to changes in traction demand. The model was assessed experimentally on a straight and level section of track using four controlled configurations: the locomotive operating alone and the locomotive pulling one, two, and three additional wagons. Fuel-cell voltage measurements from repeated runs were processed and compared with the corresponding simulation outputs, with the main assessment focused on steady-state behaviour and a separate, more limited assessment of startup transients.

The results show that the model reproduces the expected relationship between increased mechanical loading and reduced fuel-cell voltage. Promising results were obtained across the tested configurations, with the steady-state Mean Absolute Error remaining below 1% for the heaviest configuration. The larger differences occurred during startup, where the model did not fully reproduce the effects of mechanical stiction, fast electrical buffering, and other short-timescale hardware dynamics. These differences define the current limits of the model rather than invalidating its steady-state use. Within the tested G-scale operating envelope, the resulting model provides a practical platform for analysing load-dependent fuel-cell voltage behaviour and establishes a foundation for future work involving higher-rate transient measurements, current and power validation, battery hybridisation, regenerative braking, and Energy Management Strategy development.

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List of Nomenclatures

Abbreviations

Abbreviation	Meaning
AC	Alternating current
CCM	Continuous conduction mode
CSV	Comma-separated values
DC	Direct current
DC-DC	Direct-current to direct-current converter
EMS	Energy Management Strategy
G-scale	Garden-scale model railway
LTD	Longitudinal train dynamics
MAE	Mean Absolute Error
OCV	Open-circuit voltage
PEMFC	Proton Exchange Membrane Fuel Cell
PWM	Pulse-width modulation
RMSE	Root Mean Square Error
ZOH	Zero-Order Hold

Symbols and Variables

Symbol	Meaning	Unit
A	Tafel slope parameter	V
A, B, C	Davis resistance coefficients	N, Nsm^{-1}, Ns^2m^{-2}
E_b	Motor back-electromotive force	V
E_n	Nernst open-circuit voltage	V
F_D	Davis resistance force	N
F_{drag}	Total modelled drag force	N
G	Gear ratio	dimensionless
I_m	Motor armature current	A
J_{eq}	Equivalent inertia reflected to the motor shaft	$Kg.m^2$
K_e	Motor back-EMF constant	$V.s.rad^{-1}$
K_t	Motor torque constant	$N.m.A^{-1}$
L	Motor armature inductance	H
m_{train}	Total train mass	kg
p_{air}	Air pressure supplied to the fuel-cell model	Bar
p_{fuel}	Hydrogen pressure supplied to the fuel-cell model	Bar
q_{air}	Air flow rate supplied to the fuel-cell model	$cm^3.s^{-1}$
q_{fuel}	Hydrogen flow rate supplied to the fuel-cell model	$cm^3.s^{-1}$
r	Wheel radius	m
r_i	Fuel-cell internal resistance	Ω
R_m	Motor armature resistance	Ω
t	Time	s
v	Linear train speed	$M.s^{-1}$

V_{act}	Fuel-cell activation-loss voltage	V
V_{conc}	Fuel-cell concentration-loss voltage	V
V_{fc}	Fuel-cell terminal voltage	V
V_m	Motor terminal voltage	V
V_{ohm}	Fuel-cell ohmic-loss voltage	V
V_{track}	Regulated track-side voltage	V
ω_m	Motor angular velocity	$rad.s^{-1}$
τ_{load}	Load torque opposing motor rotation	$N.m$
τ_m	Motor torque	$N.m$
η	Drivetrain efficiency	dimensionless
i	Fuel-cell current density or current, according to context	$A.m^{-2}$ or A
i_0	Exchange current density	$A.m^{-2}$

Key Equations

Equation	Description
$F_D = A + B \cdot v + C \cdot v^2$	Davis-type representation of train resistance.
$V_m = K_e \omega m + R_m I_m + L \frac{dI_m}{dt}$	DC motor armature voltage balance.
$T_m = K_t I_m$	Relationship between motor current and motor torque.
$T_{load} = \frac{-F_{drag} r}{G \eta}$	Conversion of linear drag force into opposing motor-shaft torque.
$J_{eq} = m_{train} \left(\frac{r}{G} \right)^2$	Equivalent train inertia reflected to the motor shaft.
$V_{fc} = E_n - V_{act} - V_{ohm} - V_{conc}$	Reduced PEMFC terminal-voltage model.
$V_{act} = A \ln \left(\frac{i}{i_0} \right)$	Activation-loss relationship used for the Tafel-slope parameter.
$V_{ohm} = i \cdot r_i$	Ohmic voltage-loss relationship used for the internal-resistance parameter.

Data Analysis Terms

Term	Meaning
0.5 s Min-Binning	Extraction of the minimum simulated voltage within each 0.5-second comparison interval.
Transient state	The interval $t \leq 2.5$ s, selected to represent the main startup voltage transition.
Steady state	The interval $t > 2.5$ s, used for the primary load-dependent voltage assessment.
Config 1	Locomotive only.
Config 2	Locomotive plus one wagon.
Config 3	Locomotive plus two wagons.
Config 4	Locomotive plus three wagons.

1. Introduction

1.1 Project Motivation

The primary motivation for this project was to develop and validate a more physically representative model of the hydrogen-powered G-scale train used in the experimental investigation. The inherited baseline model already provided a useful mathematical description of the train's mechanical behaviour, including mass, tractive effort, acceleration, and resistance. However, its electrical power supply was represented by a constant voltage block rather than by the hydrogen fuel cell used in the physical system.

This project therefore focused on extending the existing mechanical model into a coupled hydrogen fuel-cell and railway traction model. This project was not just to simply replace the voltage source block, but also to introduce one of the main physical elements of the experimental system and examine how mechanical loading influences the electrical source. Unlike an ideal voltage source, a fuel cell has a load-dependent terminal voltage and can exhibit voltage changes when the traction motors demand different levels of current [1], [2]. Representing this behaviour allows the effect of adding wagons to be observed not only as a mechanical change, but also as a corresponding change in simulated fuel-cell voltage.

1.2 The Role of Scale-Model testing

Validating electromechanical powertrains on full-scale locomotives is very expensive and logistically complex. G-scale model trains provide a controlled and cost-effective means for testing the coupling between mechanical loading and electrical source behaviour. Scaled railway studies support this use when the model scope and scaling assumptions are stated explicitly [3], [4], [5]. The fundamental relationships between force, inertia, motor torque, and electrical loading remain useful at reduced scale, but a G-scale model is not dynamically identical to a full-size railway vehicle. Differences in friction, gearing, motor characteristics, thermal behaviour, and electrical time constants limit direct numerical application. The present project therefore uses the scale train to validate the model within its own operating envelope.

1.3 Aims and Objectives

The core objective of this project is to bridge the gap between abstract mathematical theory and physical reality by “electrifying” an existing baseline mathematical model. The specific aims are:

- 1. Architectural Migration:** To replace the baseline Simulink model's idealised, constant mathematical power supply block with a dynamic, physical Simscape Hydrogen Fuel Cell block.
- 2. Electromechanical Coupling:** To ensure that physical changes in the mechanical domain, such as adding wagon mass, produce the corresponding electrical loading and fuel-cell voltage response.
- 3. Sequential Calibration:** To calibrate the model and estimate the permitted fuel-cell parameters (the Tafel slope and internal resistance), by manually varying them while holding the remaining baseline parameters fixed.
- 4. Validation:** To compare the resulting simulation outputs with real-world hardware data across the four mass configurations, with primary emphasis on steady-state voltage agreement.

1.4 Thesis Outline

In the rest of this report, Chapter 2 provides a literature review on longitudinal train dynamics and fuel cell modelling. Chapter 3 analyses the limitations of the inherited mathematical baseline model. Chapter 4 details the architectural migration into the hybrid physical-mathematical Simscape environment. Chapter 5 outlines the physical G-scale hardware setup and the data acquisition and processing pipeline. Chapter 6 details the parameter estimation methodology and presents the statistical validation of the physical model against the hardware data. Finally, Chapter 7 summarises the project's contributions and outlines areas for future work and EMS optimisation.

2. Literature Review

2.1 Scope and positioning of the present project

Hydrogen traction is now treated as a serious option for rail routes where full electrification is hard to justify. The literature has converged on a hybrid architecture in which a proton exchange membrane fuel cell supplies average power, while a battery or other storage device supplies fast transients, accepts regenerative braking energy, and buffers the fuel cell from sharp load changes [1]. This broader system context is important for the present project, but the implemented system is narrower: a PEMFC power source, a regulated DC-DC converter, DC traction motors, and a mechanically loaded G-scale locomotive.

The experimental validation reported here used a straight, level section of track and four mass configurations: the locomotive alone and the locomotive pulling one, two, and three additional wagons. The main measured quantity was fuel-cell voltage sampled at 0.5-second intervals. Curves, gradients, odometry, regenerative braking, battery state of charge, and hydrogen consumption are relevant extensions of the wider hydrogen-rail problem, but they were not the primary observables in this validation project report. They are therefore treated as background and future directions rather than as measurements already completed.

The project sits at the boundary between longitudinal train dynamics and physical fuel-cell powertrain modelling. Its contribution is not a full-scale prediction of railway operation. It is the validation of a coupled G-scale model in which changes in mechanical loading produce corresponding changes in electrical source behaviour or in other words voltage at the Fuel Cell.

2.2 Main threads of prior work

The relevant literature can be grouped into five connected threads:

- Hydrogen and hybrid rail demonstrators establish that fuel-cell traction can be implemented on vehicles, mini-trains, and test benches, and show the types of operating data needed to assess the integrated system [6], [7], [8], [9], [10], [11].
- Fuel-cell and energy-management modelling studies provide system-level fuel-cell representations, component-sizing methods, power-distribution strategies, and model-to-test-bench comparisons [12], [13], [14], [15].
- Longitudinal train-dynamics studies describe the relationship between mass, resistance, acceleration, tractive effort, and route conditions [8], [16].
- Scaled rail and roller-rig studies show how reduced-scale platforms can be instrumented and compared with computational models [3], [4], [5], [17], [18].
- Model-validation studies frame accuracy in relation to the purpose, scope, observables, and uncertainties of the model rather than an unrealistic demand for complete physical detail [16].

The present project combines these concepts in a more focused way. It uses a scaled railway platform, but the validation target is not general vehicle dynamics alone. It is the coupled response from added mechanical mass through the motor and converter to the PEMFC voltage.

2.3 Hydrogen-hybrid traction as the relevant system background

Hydrogen rail studies consistently treat the fuel cell as a relatively steady or slowly varying source and the battery or other energy-storage system as the device that covers peaks, accepts regenerative braking energy, and protects the fuel cell from rapid demand swings [1], [9]. The common full-system topology contains a fuel-cell converter, an energy-storage converter, a shared DC bus, an inverter, and the traction

motors [1]. Full vehicle demonstrations show why these elements must be considered together. HydroFLEX testing, for example, revealed that rapid traction transients could trigger battery protection limits and require changes to both hardware thresholds and control logic [10].

This architecture matters for model development because the train is not governed by a single energy source. Vehicle motion, route resistance, fuel-cell voltage, storage behaviour, and power-split control interact. However, the present project deliberately isolates the fuel-cell and mechanical-load relationship before introducing a battery or a full energy-management strategy. The validated PEMFC-converter-motor model can therefore be understood as a foundation for later hybridisation rather than as a complete hybrid train model.

Hydrogen rail case studies also show the value of measured operating data. Test-vehicle work in Japan measured fuel-cell and battery behaviour over repeated railway duty cycles and used the resulting energy flows to evaluate efficiency and fuel consumption [9]. The HydroFLEX programme combined simulation, hardware integration, track testing, and mainline data collection to tune the powertrain and its control limits [10]. These studies support the project's decision to validate an integrated physical model against hardware data rather than relying on component specifications alone.

2.4 Longitudinal train dynamics and mass-dependent traction demand

A longitudinal train model converts route conditions, train mass, speed, and acceleration into the tractive effort required at the wheel-rail interface. A common resistance representation is the Davis-type relation

$$F_R = A + Bv + Cv^2$$

where the terms represent approximately constant, speed-dependent, and aerodynamic components respectively. Gradient(Track inclination) and curve resistance can be added when the route requires them. The relative importance of these terms changes with speed and operating condition, while mass directly affects acceleration and the mass-dependent part of rolling resistance [16].

This framework explains why the four mass configurations used in the present project are informative even though the track was straight and level. Removing gradients and curves reduces the number of varying external forces, so the effect of added wagon mass can be observed more clearly. Adding wagons increases both the translational inertia of the train and the traction force needed to overcome resistance. The resulting change in motor current and fuel-cell voltage provides a direct test of whether the mechanical and electrical parts of the model are properly coupled.

The mechanical model also needs to account for rotational inertia reflected through the gearing. For a wheel radius r , gear ratio G , and train mass m , the approximate mass contribution reflected to the motor shaft is

$$J_{eq} = m \left(\frac{r}{G} \right)^2$$

while the resistive force can be converted to a shaft load in presence of drivetrain efficiency value:

$$\tau_{load} = \frac{(-F_{res} \cdot r)}{(G \cdot \eta)}$$

These relations are central to the project's migration from a functional traction-force calculation to a physical rotational network. They provide the mechanical path by which the locomotive and wagon mass becomes motor torque demand and, ultimately, electrical loading.

The literature also warns against treating a single deterministic train simulation as universally valid. Train models contain uncertainty in mass, resistance coefficients, route data, friction, and drivetrain behaviour. The appropriate model complexity depends on the question being asked [16]. For the present short consist and flat-track experiment, a compact longitudinal model is appropriate, provided its assumptions and validation limits are stated clearly.

2.5 Multi-domain physical modelling and electromechanical coupling

The inherited baseline model represented the train largely through mathematical blocks, integrators, and algebraic relationships. This is efficient and can be appropriate for a well-defined longitudinal model, but it makes the mechanical-to-electrical energy path dependent on the accuracy of manually written equations. A physical-domain model instead represents the main energy-storage and conversion elements as connected physical components. In the present case, the Simscape architecture contains the fuel-cell source, converter, motors, shaft inertia, force-to-torque conversion, and mechanical load in one coupled network.

This change is important because the electrical and mechanical domains obey linked conservation relationships. The DC motors convert electrical current into torque and generate back-EMF as their speed increases. A simplified motor representation is

$$V_m = K_e \omega_m + R_m I_m + L_m \frac{dI_m}{dt}$$

and

$$\tau_m = K_t I_m$$

where V_m is motor voltage, I_m is armature current, ω_m is motor speed, R_m and L_m are electrical parameters, and K_e and K_t are the back-EMF and torque constants. The motor torque is transferred through the gear train to the wheelset, while resistance and inertia act back on the shaft.

The project therefore does not just replace one voltage source with another. It replaces an infinite ideal supply with a source whose voltage changes when the mechanically coupled system demands more current. This is the mechanism that allows added mass to affect the simulated fuel-cell voltage. The model's good steady-state agreement is consequently evidence that the combined representation of mass, resistance, gearing, motor torque, and fuel-cell loading is internally consistent for the tested operating range.

The model should nevertheless be described as a physical approximation rather than a complete representation of every component. The average-value converter, simplified fuel-cell dynamics, and faultless shaft elements reduce computational cost but ignore some high-frequency and transient phenomena. The value of the physical architecture is therefore that it implements the main domain relationships, not that it removes all model-form uncertainty.

2.6 PEMFC polarisation and low-order dynamic modelling

A PEMFC's voltage is lower than its equilibrium value because of activation, ohmic, and concentration losses. A system-level expression is

$$V_{fc} = E_N - V_{act} - V_{ohm} - V_{conc}$$

where E_N is the Nernst voltage and the remaining terms represent the main voltage losses. This structure provides a useful bridge between electrochemical behaviour and a traction-chain model. It explains why the voltage falls as current rises and why the shape of the polarisation curve is not purely linear [1].

Polarisation-curve studies show that parameters such as the Tafel slope, exchange-current behaviour, internal resistance, temperature, and reactant conditions influence the fitted voltage-current relationship [19]. Dynamic PEMFC models extend this steady-state picture by including variable-load behaviour, gas-flow effects, temperature effects, water management, and startup phenomena [2]. In railway applications, however, full electrochemical detail is often unavailable or unnecessary. A model intended for system simulation must balance the ability to be represented physically with the need for identifiable parameters and manageable computation [13].

This trade-off is directly relevant to the present project. The Horizon H-60 fuel cell is a commercial component whose internal electrochemical parameters are not fully observable. The model therefore uses a reduced representation calibrated against the measured operating behaviour. The Tafel-slope parameter controls the curvature of the low-current voltage response, while the internal resistance controls the approximately linear voltage spacing between steady-state mass configurations. This approach is consistent with system-level identification methods that use mission or operating-profile measurements to obtain a practical traction-chain model [13].

The calibration should not be interpreted as proof that the used parameters are unique physical properties of the stack. Different combinations of electrochemical, converter, motor, and load parameters can sometimes produce similar terminal-voltage curves. The tuned model is therefore best understood as an effective representation of the tested system and operating range.

2.7 Fuel-cell transient response and model fidelity

The distinction between steady-state and transient behaviour is central to the present results. A PEMFC can follow a changing electrical demand through several interacting mechanisms, including electrochemical reaction kinetics, ohmic losses, reactant transport, gas-flow dynamics, water management, temperature, and electrical double-layer effects [2]. These mechanisms operate on different time scales. A low-order polarisation model may reproduce the settled voltage-current relationship while missing the exact shape and depth of a rapid startup voltage dip.

The observed discrepancies in the project are consistent with this model-fidelity issue. In the lightest configuration, the hardware exhibits a deeper initial dip relative to model and that is possibly associated with breakaway friction and stiction in the locomotive drivetrain. In the heaviest configuration, the model predicts a deeper dip than the hardware, indicating that the real source and converter contain transient buffering that is not represented fully in the model. The latter should be discussed carefully: fuel-cell double-layer capacitance, converter inductance, output capacitance, and control-loop dynamics can all influence the measured terminal response and should not be treated as interchangeable effects.

This distinction supports the decision to report transient and steady-state errors separately. A model may be highly useful for predicting the effect of sustained mechanical loading while remaining unsuitable for detailed prediction of initial behaviour. The literature on model fitness for purpose supports judging the model against the outputs and time scales for which it is intended to be used [16].

2.8 Demonstrator evidence closest to the present project

Hydrogen rail research has explored several physical scales and system configurations. Early work demonstrated small PEMFC-powered locomotive concepts using onboard hydrogen storage [6]. The hybrid PEMFC mini-train developed by Hsiao and colleagues provided long-duration operating experience and showed that a fuel-cell and lead-acid-battery system could supply stable traction power in a small vehicle [7].

The Hydrogen Pioneer project provides a particularly relevant precedent because it combined a hydrogen fuel-cell power plant with batteries and measured resistance to motion, operating speed,

efficiency, and power sharing [8]. These measurements link vehicle mechanics to source-side performance in a way that is close to the logic of the present experiment.

Larger railway demonstrators provide evidence of the additional complexity that appears when the same basic architecture is scaled up. Japanese hybrid test vehicles measured fuel-cell output, battery behaviour, energy flows, and fuel consumption over repeated test-track operation [9]. HydroFLEX showed that rapid traction transients can trigger protection limits and require coordinated changes to hardware thresholds and control logic [10]. More recent bench work has also used route-based inputs to characterise a scaled fuel-cell and battery system [11].

Taken together, these studies support three conclusions. First, hydrogen rail traction has been investigated across model, mini-train, bench, prototype, and mainline platforms. Second, measured integrated behaviour is essential because component specifications alone do not capture the response of the complete vehicle. Third, the present project occupies a narrower but useful position by isolating the relationship between added mechanical mass and fuel-cell voltage before adding battery buffering and EMS control.

2.9 Scaled validation and transferability

Roller-rig and scaled-vehicle studies have long been used to validate railway vehicle models when full-scale testing is costly, risky, or impractical [4]. A 1:5 roller-rig study demonstrated correlation between measured rig behaviour and a MATLAB railway vehicle model [3]. Other work has treated the design of a scaled railroad vehicle itself as part of the model-validation problem [5], while narrow-gauge and model-railway studies have used measured speed, tractive force, resistance, and motion data to compare scaled experiments with computation [17], [18].

The important claim is not that a G-scale train is dynamically identical to a full-size railway vehicle. The stronger and more defensible claim is that a reduced-scale platform can test whether a specified model reproduces the behaviour of that platform under controlled conditions. Scaling changes wheel-rail contact, friction, gear losses, motor size, thermal behaviour, electrical time constants, and the relative importance of different resistance terms. These differences limit direct numerical extrapolation to full-size rail vehicles.

The present experiment therefore validates the coupled G-scale model within its own operating envelope. Some structural relationships such as the dependence of traction demand on mass, the conversion of force to shaft torque, and the qualitative effect of a load-dependent source, are useful beyond this particular setup. Absolute voltage levels, transient time constants, friction parameters, converter behaviour, and fuel-cell ratings are not automatically transferable. Future scale-up would require a separate similarity analysis and re-identification of full-scale component parameters.

This distinction strengthens the project. A well-bounded scale experiment can reveal whether the model architecture and coupling logic are sound before more expensive vehicle-level development is attempted. It should not be presented as a substitute for full-scale certification or full-scale performance validation.

2.10 Validation observables and fitness for purpose

The broader railway-model literature identifies speed, distance, tractive force or power, resistance, and route profile as important validation observables [16]. Model-train experiments have also shown the value of direct measurements of speed and tractive force, rather than relying only on inferred behaviour [18]. Hydrogen rail demonstrators add source-side observables such as fuel-cell output, battery contribution, DC-bus voltage, hydrogen use, and energy efficiency [8], [9], [10].

The present project validates a more limited set of observables. Its strongest evidence concerns the fuel-cell voltage response during acceleration and settled motion under four different mechanical loads.

The results therefore support claims about load-dependent source voltage and steady-state electromechanical coupling. They do not, by themselves, establish accurate hydrogen consumption, complete electrical power balance, regenerative-braking behaviour, or route-dependent resistance over curves and gradients.

This is a fitness-for-purpose question. A model intended to investigate the steady-state electrical consequences of added mass does not need the same detail as a model intended to design a battery EMS, predict regenerative braking, or certify a full-scale train. The model's scope, inputs, outputs, time resolution, and validation criteria should be stated together so that the reported accuracy is not interpreted more broadly than the evidence allows [16].

2.11 Relevance to future hybridisation and energy-management strategies

The fuel-cell-only configuration is a useful first stage for future hybridisation. Full hydrogen railway vehicles normally add a battery or supercapacitor so that the fuel cell can operate closer to an efficient and durable operating region while the storage system supplies rapid transients and absorbs regenerative energy [1]. Experimental railway studies show that the power-sharing controller and storage limits can strongly affect the measured duty-cycle efficiency and fuel consumption [9].

Energy-management studies also show that strategy performance depends on the level of detail used in the component models. Load-following, rule-based, and Pontryagin-based strategies can produce different results when battery relaxation, fuel-cell characteristics, aging, and uncertain parameters are represented differently [14], [15]. This provides a direct reason to validate the fuel-cell and traction model before using it to compare EMS approaches.

The present project does not demonstrate a battery EMS. Its contribution is to establish a physically coupled source-and-load model that can later be extended with a battery or supercapacitor, bidirectional power conversion, regenerative braking, and an energy-management controller. The validated steady-state load response provides a baseline against which future transient buffering and power-split strategies can be assessed.

2.12 What the literature implies for the present project

The literature supports the following interpretation of the project design:

- The G-scale platform is justified as a controlled experimental environment for validating a coupled railway model, but not as a direct full-scale substitute [3], [4], [5].
- The mass-variation experiment is appropriate because it isolates a clear mechanical effect and tests whether the resulting traction demand is transferred correctly into electrical source behaviour [8], [16].
- The Simscape migration is justified because the physical network makes the motor, inertia, drivetrain, converter, and fuel-cell source part of one coupled energy-conversion model.
- A reduced PEMFC representation is appropriate for system-level traction analysis when internal stack parameters are unavailable, provided its operating range and transient limitations are stated [2], [13].
- Separate transient and steady-state validation is necessary because the simplified model omits some friction, capacitance, transport, and converter dynamics [2], [16].
- Future battery and EMS work should build on the validated fuel-cell and mechanical model rather than being treated as already demonstrated by the current experiment [1], [14], [15].

2.13 Small critical gap analysis

The literature is strong, but it does not fully cover the present configuration.

First, hydrogen rail demonstrator papers usually focus on proof of concept, integrated vehicle performance, or hybrid power management [8], [9], [10]. They rarely combine these with a detailed G-scale validation of a physical PEMFC-converter-motor model under controlled mass changes.

Second, scaled rail validation papers are rich in vehicle dynamics, resistance, instrumentation, and simulation comparison [3], [4], [5], [17], [18], but they generally do not include PEMFC source dynamics. Their evidence is strongest on mechanical behaviour rather than coupled electrochemical and traction response.

Third, fuel-cell traction-chain models provide practical identification and control methods [13], [14], but they usually work from full-scale missions or bench environments rather than from a compact locomotive with a small commercial fuel cell and a low-rate voltage measurement.

The resulting gap is therefore not a lack of relevant literature. It is the lack of an integrated, project-scale study that tests whether a physical railway model can translate controlled changes in mechanical loading into realistic fuel-cell voltage behaviour. The present work addresses that gap within a clearly bounded G-scale operating envelope.

2.14 Synthesis

The literature supports the present project on both sides of its design. Hydrogen rail research provides a basis for modelling fuel-cell traction, source dynamics, power conversion, hybridisation, and energy-management requirements [1], [9], [14]. Railway validation research provides a basis for using reduced-scale platforms to compare measured and simulated behaviour when the scaling assumptions and model boundaries are explicit [3], [4], [5], [17]. Longitudinal train-dynamics literature explains how mass, resistance, acceleration, and drivetrain behaviour generate the traction demand that ultimately appears as electrical loading [16].

The main research opportunity lies in bringing these threads together around a G-scale locomotive and wagons. The present project does this by validating the steady-state load-dependent voltage response of a PEMFC-powered electromechanical model across four mass configurations, while identifying transient limitations associated with stiction and missing electrical buffering. The resulting model is best viewed as a validated foundation for later studies of batteries, supercapacitors, regenerative braking, and EMS optimisation rather than as a complete full-scale hydrogen railway simulation.

3. Mathematical Model of the G-Scale Model Train

3.1 The Baseline Model:

At its core, this baseline Simulink model acts as an accurate mathematical representation of the model train, establishing the mechanical load that the future electrical system must eventually power. To capture Longitudinal Train Dynamics (LTD), the architecture was built with two distinct testing concepts: a Combined Mass System and a Distributed Mass System. The Combined Mass System represents the train as a single, rigid point-mass, relying on a custom MATLAB function (*CombinedMassModel*) to determine the fundamental acceleration ($\ddot{x}$) using Newton's Second Law of Motion:

$$\ddot{x} = \frac{F_{trac} - R_D - F_G}{m}$$

From this calculated acceleration, standard Simulink Integrator blocks sequentially convert the data into dynamic speed ($\dot{x}$) and track position (x). However, to truly stress-test a powertrain, one must account for the physical ripple and dynamic feedback of individual rolling stock. The Distributed Mass System expands the baseline by calculating the physical push and pull between the locomotive and up to three subsequent wagons. The masses of the vehicles are defined discretely (e.g., $m = [10.5 ; 8 ; 8 ; 8]$ for the locomotive and three wagons). The model simulates the inter-wagon couplers as a mass-spring-damper system, calculating the coupler force (F_c) between adjacent cars using a specified spring stiffness (k) and a damping coefficient (c):

$$F_c = k(\Delta x - d) + c(\Delta \dot{x})$$

3.1.1 Tractive Effort and Aerodynamic/Rolling Resistance

To make the virtual train move, the baseline model translates electrical characteristics into physical mechanical force. The DC motors act as the bridge between the electrical supply and the track. By taking workspace parameters such as armature resistance (R_m), inductance (L), and the back-EMF constant (K_e), the model calculates the raw motor output. This is passed through a "Traction Force Calculation" gain block, which applies the motor's torque constant (K_t), gearbox efficiency (η), and the physical wheel radius (r) to determine the final tractive effort (F_{trac}) applied to the rails. However, this forward movement is constantly opposed by environmental drag. The model incorporates a dedicated "DavisResistance" subsystem that continuously calculates the opposing aerodynamic and rolling drag (R_D) dynamically based on the train's current velocity (Figure 1).

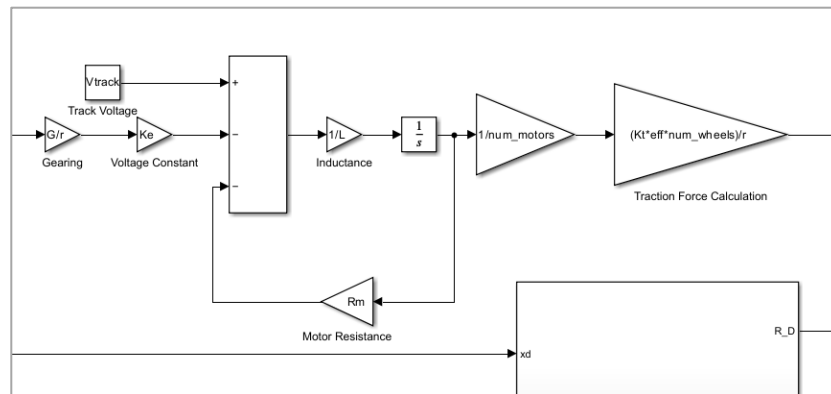


Figure 1 Baseline Mathematical Model Dynamics in Simulink

3.2 The Baseline Electrical Representation

The inherited baseline model was developed primarily to represent the mechanical behaviour of the train. Its electrical supply was therefore represented by a constant "Track Voltage" block set to 14V (Figure 1). This was a deliberate idealisation within the scope of the original mathematical model, allowing the mechanical equations to be studied without modelling the detailed behaviour of an onboard energy source.

The constant block did not represent the hydrogen fuel cell used in the physical train. Consequently, it could not reproduce the source-side voltage response associated with changes in motor current and mechanical loading. A real fuel cell has voltage and current characteristics that depend on its operating conditions and electrical demand, including activation and ohmic losses [1], [2]. Replacing the constant block was therefore necessary to represent the actual hydrogen power source and to make comparison with the physical fuel-cell voltage measurements meaningful.

4. System Modelling and Dynamics

4.1 Introduction and Architecture Overview

The initial baseline simulation for the scale model train relied on pure mathematical abstractions in Simulink, using transfer functions and integrators to represent the train's dynamics. While computationally simple, pure mathematical models can obscure physical scaling errors and make the energy path between electrical and mechanical domains dependent on manually written equations.

To support comparison with experimental data, the architecture was migrated to a hybrid physical-mathematical modelling environment using Simscape. The migrated model represents the main mechanical and electrical elements as connected physical networks, allowing their coupled behaviour to be resolved by the physical solver. The model is logically partitioned into four distinct subsystems: the Fuel Cell power source, the Power Regulation circuit (Buck Converter), the Electromechanical Drive, and the Kinematic & Drag calculation blocks. (Figure 2)

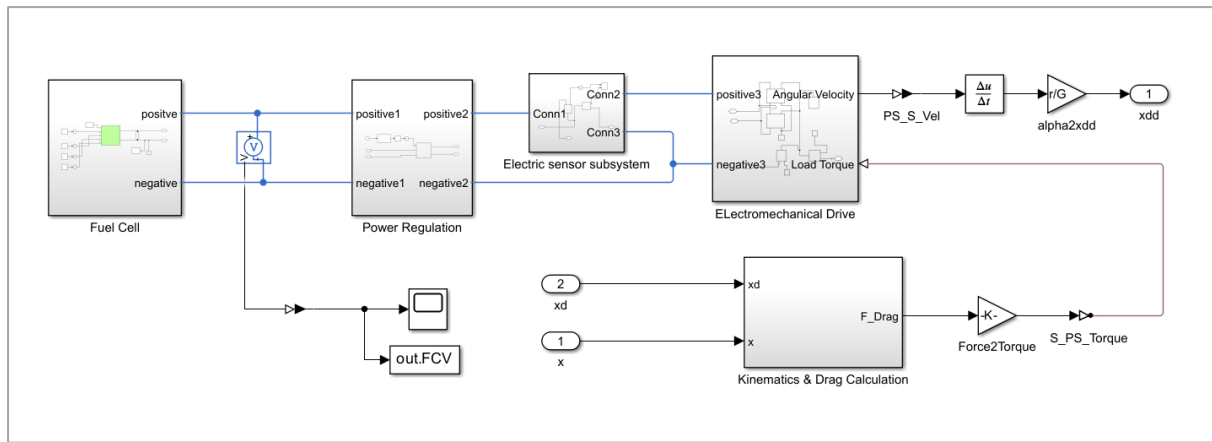


Figure 2 Simulink Model of Hydrogen Fuel Cell powered G-Scale model train

4.2 The Fuel Cell Subsystem

The primary power source of the scale model train is a hydrogen fuel cell. Because a fuel cell's voltage depends on current load and is affected by activation, ohmic, and concentration losses, it was modelled using Simscape's physical Fuel Cell block rather than an ideal DC voltage source [1], [2], [19].

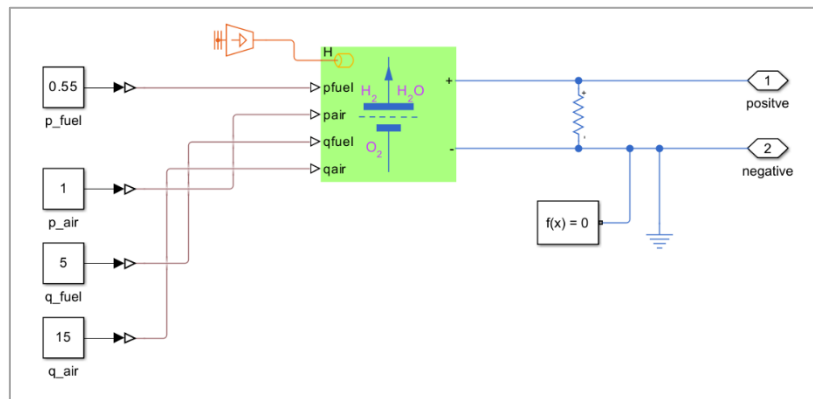


Figure 3 Fuel Cell Subsystem

4.2.1 Input Parameters and Operating Conditions

To ensure the fuel cell accurately reflects the physical environment of the scale model, several physical parameters are fed into the block via Simulink-PS (Physical Signal) converters (Figure 3):

- **Temperature:** An absolute temperature source is provided to dictate the operating temperature of the cell as 20° C, which directly impacts the exchange current density.
- **Gas Pressures (p_{fuel} , p_{air}):** The supply pressures for the hydrogen fuel and ambient air (oxygen) are provided as constant inputs of values 0.55 bar and 1 bar respectively.
- **Flow Rates (q_{fuel} , q_{air}):** The reactant flow rates are specified to ensure the model correctly calculates Nernst potentials and concentration losses at high loads.

4.2.2 The Initialisation Resistor

A common challenge in Simscape when connecting non-linear voltage sources directly to switching converters or inductors is the occurrence of algebraic loops and initialisation solver errors. To resolve this, a 1000 Ω resistor was connected in parallel across the fuel cell terminals. This parallel resistor serves primarily as a numerical shunt:

- **Solver Stability:** It provides a finite resistance to ground, helping stabilise the initial solver state matrix and avoid open-circuit initialisation errors.
- **Model Boundary:** It provides a small continuous electrical path in the simulation, but it was not independently measured and should not be interpreted as a direct measurement of fuel crossover.

4.3 Power Regulation

In the experimental setup, a buck converter steps down the highly variable fuel cell voltage (15V - 19V) to a stable 14V track voltage.

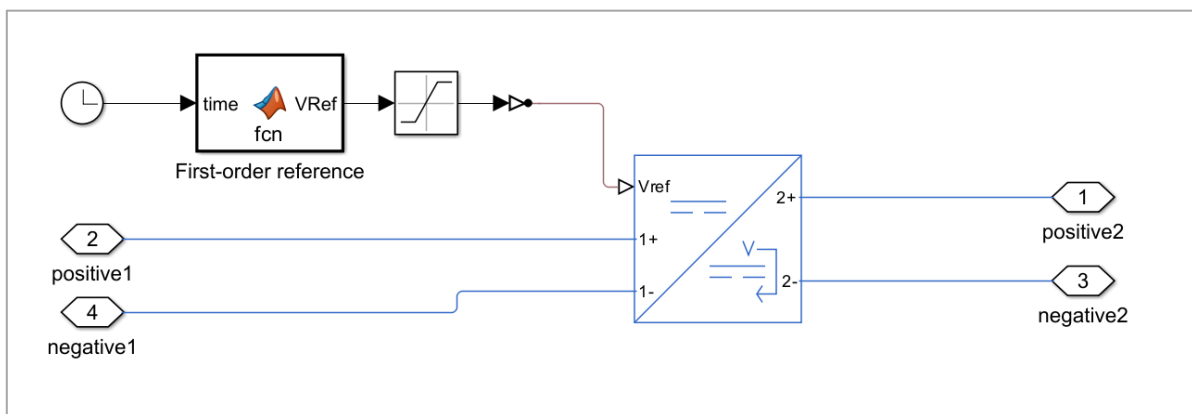


Figure 4 Power Regulation Subsystem

The output track voltage is regulated using a custom MATLAB Function block. This block operates as an open loop controller. It takes the simulation Clock time and produces a first-order response with the maximum limit set to 14V in the saturation block that follows. This simulates a voltage starting at 0 V, rising and eventually settling at 14V. To replicate a buck converter, this signal is then fed into an Average-Value DC-DC Converter block. Unlike standard mathematical gains, the Average-Value model natively simulates the voltage reference-dependent power transfer. (Figure 4)

4.4 Electromechanical Conversion (Motors & Loads)

The transition between the electrical track power and the physical motion of the train is handled by the mechanical rotational network.

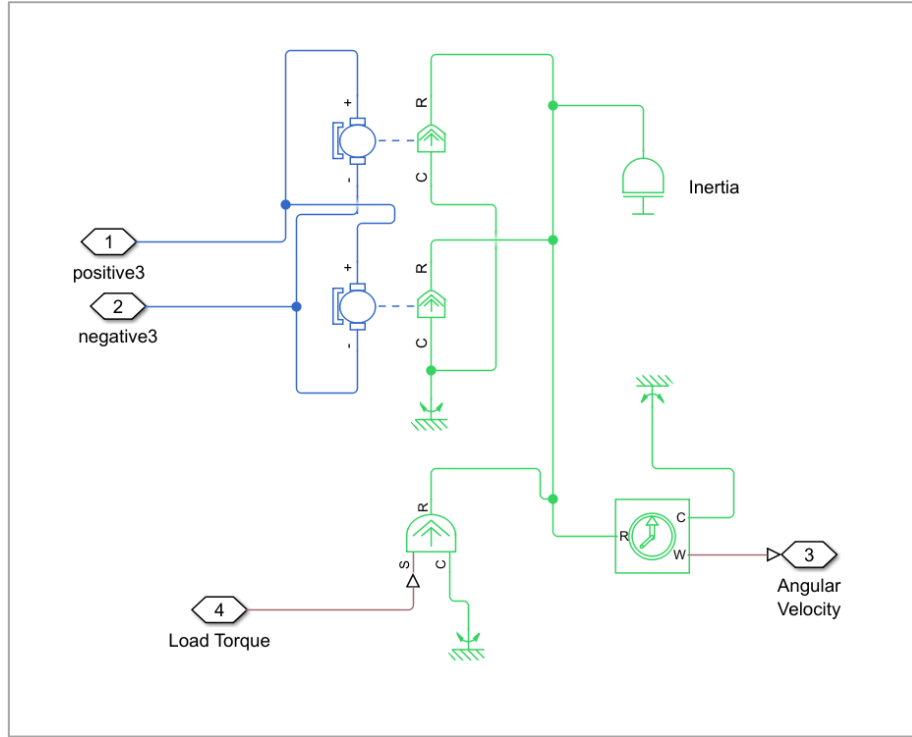


Figure 5 Electromechanical Drive Subsystem

Two parallel Simscape DC Motor blocks (Figure 5) replace the custom integrator math of the baseline model. These blocks automatically solve the standard DC motor armature equation:

$$V_{track} - E_b = I_m \cdot R_m + L \cdot \frac{dI_m}{dt}$$

Where back-EMF (E_b) is proportional to the motor's angular velocity (ω_m) via the voltage constant (K_e), and torque (τ_m) is directly proportional to armature current (I_m) via the torque constant (K_t). The two motors output their combined torque directly to a shared physical rotational shaft.

To integrate the electromechanical system within Simscape, the mechanical load is connected to the physical motors through the corresponding force-to-torque relationship. The linear drag of the train (F_{drag}) is translated into a resistive rotational load via the Force2Torque gain block, using the modelled gear ratio ($G=30$) and drivetrain efficiency (eff) (Figure 2):

$$\tau_{load} = -F_{drag} \cdot \left(\frac{r}{G \cdot eff} \right)$$

This conversion allows the motors to operate with the modelled mechanical advantage and establishes the intended relationship between train loading and simulated current demand.

The resistive load torque calculated above is applied directly against the motors using an Ideal Torque Source connected to the mechanical shaft. To ensure accurate acceleration, an Inertia block is also connected to the shaft. Rather than mathematically integrating force, the mass of the train is reflected through the gearbox to the motor shaft using the pure kinetic energy equivalence formula:

$$J_{eq} = m_{train} \cdot \left(\frac{r}{G}\right)^2$$

4.5 Kinematics and Drag Calculation

The final section of the model handles the calculation of the train's speed, distance, and total environmental drag using mathematical feedback loops connected to the physical network.

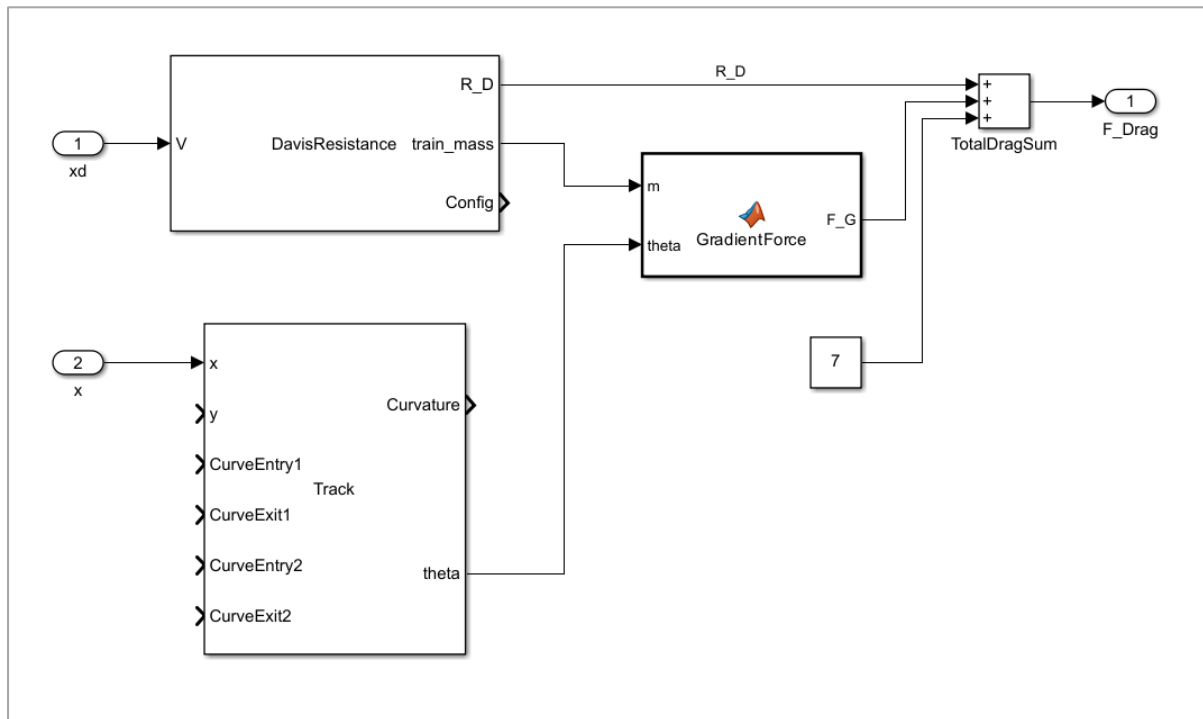


Figure 6 Kinematics and Drag Calculation Subsystem

4.5.1 Sensor Feedback and Kinematics (x, xd, xdd)

An Ideal Rotational Motion Sensor is attached to the Simscape mechanical shaft to physically measure the motor's angular velocity (ω_m). This signal is passed back into the Simulink mathematical domain and processed to yield the standard kinematics (Figure 2):

- Acceleration (xdd): The angular velocity is passed through a Derivative block and then a gain block of value r/G to get linear acceleration.
- Velocity (xd): The linear acceleration is then passed through a pure integrator to get linear velocity v .
- Distance (x): Then, velocity v is passed through a pure Integrator block to calculate total distance traveled.

4.5.2 Translating Baseline Traction Dynamics

In the original baseline model, the vehicle's electromechanical dynamics were modelled using pure functional equations. The simulation explicitly calculated a linear Traction Force from the motor current, which was then opposed by the standard environmental drag (Davis Resistance and Gradient Force). Within those baseline Traction Force equations, certain dynamic load compensations were inherently mathematically bundled.

When migrating to the Simscape architecture, the electromechanical dynamics are handled natively by physical components. The DC Motor blocks physically convert electrical power into rotational torque, eliminating the need to explicitly calculate Traction Force with functional mathematical blocks. However, because the implicit mathematical compensations bundled within the baseline's Traction Force equations were removed in this migration, the resulting net force balance shifted. To bridge this gap, a constant 7N force block was added to the total drag sum after some tuning. This block is treated as an empirical model-form correction required by the migration rather than as a separately measured physical resistance.

5. Physical Hardware Setup and Data Acquisition

5.1 G-Scale Locomotive and Track Infrastructure Setup

The physical experimentation was conducted using commercially available G-Scale model railway components. The primary traction unit was a PIKO 37584 EBS BR232 Diesel Locomotive VI (Figure 7).



Figure 7 Locomotive G-Scale Model

To systematically scale the mechanical load during testing, up to three unpowered wagons were utilised, specifically the PIKO 37012 DB Eaos Gondola IV (Figure 8).



Figure 8 Wagon G-Scale Model

To ensure strict parity between the physical hardware and the mathematical simulation parameters, precision weights were added to the wagons so that each matched its respective mass parameter defined in the Simulink model.

Testing was conducted on a straight, flat section of G-scale track with no curves or gradients. Each test cycle consisted of an approximate 10-second run profile where the train accelerated from a standstill into steady-state motion before coming to a stop.

5.2 Horizon H-60 Fuel Cell Specifications and Integration

The entire powertrain was driven by a Horizon Educational H-60 PEM Fuel Cell (Figure 9), rated for 60W of continuous power. Because a raw fuel cell's output voltage fluctuates significantly based on the current load (ranging from $\sim 19.1\text{V}$ at open-circuit down to $\sim 12\text{V}$ under heavy load), the manufacturer-provided buck converter was integrated between the fuel cell and the track. This converter successfully clamped the highly variable upstream voltage to the stable 14V required by the locomotive's DC motors.



Figure 9 Horizon H-60 Hydrogen Fuel Cell

5.3 Sensor Architecture and Data Acquisition

While the original baseline architecture utilised a simple onboard accelerometer to estimate velocity and odometry, this validation phase focused on the fuel-cell voltage response to varying physical masses. Therefore, the core measured quantity was the load-dependent fuel-cell voltage during transient and steady-state operation.

Fuel-cell voltage was monitored using an HP Agilent Keysight 34401A digital multimeter and the associated HP Agilent Keysight 34401A Control and Data Logging Software (link is provided in Appendix). A critical hardware limitation of this setup was its fixed polling rate; the system was restricted to logging data in samples that are 0.5 seconds apart. This low-resolution constraint limited the ability to resolve high-speed transient dynamics.

5.4 Experimental Design

To compare the model and hardware across controlled mechanical loads, the physical train was subjected to four distinct test configurations:

- Config 1: Locomotive Only
- Config 2: Locomotive + 1 Wagon
- Config 3: Locomotive + 2 Wagons
- Config 4: Locomotive + 3 Wagons

Multiple 10-second run profiles were recorded for each configuration to ensure statistical reliability.

5.5 Data Processing Methodology

The following details the processing pipeline developed to clean, align, and average the experimental Fuel Cell Voltage data, as well as the methodology for accurately comparing it against the physical model.

5.5.1 Raw Data Characteristics & Challenges

- **Format:** Data was collected manually and saved in CSV format, recording Fuel Cell Voltage at 0.5-second intervals.
- **Configurations:** Data was separated into four folders: Locomotive Only, Locomotive + 1 Wagon, Locomotive + 2 Wagons, and Locomotive + 3 Wagons (with roughly 10 runs per configuration).
- **Physical Phases:**
A typical run consisted of:
 1. Open Circuit Voltage (OCV) at idle (~18.1V)
 2. An abrupt initial voltage dip caused by the sudden starting current draw (dropping to ~13-14V).
 3. A recovery to a steady-state moving voltage (~15V).
 4. A final rise back to OCV when the train halts.
- **Data Artifacts:**
Because recording was started manually, the actual train start time varied across files, which meant that the arrays were misaligned and of varying lengths. Also, Manual probe contact during the moving phase resulted in occasional voltage spikes due to probe slips, registering as brief 1-2 sample positive spikes (jumping to 16-17V) during the steady state.

5.5.2 Alignment & Filtering Pipeline

To create a clean, representative average for each configuration, a custom pipeline was developed:

A. Robust Trigger Detection- To align the runs, the exact start and end points of the train's movement needed to be identified. A simple threshold drop could be falsely triggered by an early probe slip. To solve that, the pipeline identifies all regions where the voltage drops below 17.5V. It then calculates the lengths of these continuous regions and selects the longest contiguous segment. This isolates the most likely train moving phase and identifies the start dip and final stop.

B. Selective Median Filtering (Preserving the Dip)- Initially, applying a global median filter (medfilt1 with a window size of 5) successfully removed the probe slip spikes, but it completely erased the physical initial current dip because the dip was only 1-2 samples long. So, to solve that, the pipeline selectively applies the median filter only to the steady-state part of the curve. By using the trigger detection indices, the script intentionally skips the first two samples of the moving phase (preserving the true physical voltage dip) and filters the rest of the array.

C. Dynamic Resampling & Averaging- Because the train runs took slightly different amounts of time, the length of the steady-state phase varied across the CSV files. To address that, for each configuration, the script found the maximum length of the steady-state segment across all runs. It then used linear interpolation (interp1) to stretch/resample the moving phase of every run to match this maximum length. The idle OCV segments before and after were padded so every array was exactly the same length. With the start, steady-state, and stop events temporally aligned, the arrays were averaged into a representative configuration curve.

5.5.3 Data Trimming for Simulation Comparison

For a direct 1:1 comparison with the simulation (which assumes the train starts immediately at $t=0$), the extensive idle data was unnecessary. A secondary script was written to isolate the relevant data: It reads the averaged files and locates the start trigger (drop below 17.5V). It trims the beginning of the file to retain exactly one sample of OCV right before the initial dip. It completely removes the trailing OCV data, stopping exactly at the final steady-state sample before the voltage rises.

5.5.4 Simulation Output Processing (Min-Binning)

When plotting the simulation output against the trimmed experimental data, a critical sampling issue was discovered.

- **The Zero-Order Hold (ZOH) Problem:** Sampling the simulation with a 0.5s interval occasionally missed the absolute minimum of the initial voltage dip because the transient trough fell between the 0.5s recording ticks.
- **Continuous Binning Solution:** The high-resolution continuous simulation data was exported directly. A comparison script dynamically down-samples this continuous data. It explicitly extracts the exact OCV value at $t=0$ for the first sample to match the experimental format. For all subsequent samples, it analyses the continuous data within that time window (0.5 seconds) and extracts the absolute `min()` voltage. This approach retains the minimum simulated voltage within each 0.5-second bin. It improves consistency when comparing the continuous simulation with the sparsely sampled experimental signal, although it cannot recover experimental transient information that was not recorded.

6. Parameter Estimation and Model Validation

6.1 Parameter origin and sequential calibration strategy

The model contains many parameters inherited from the baseline model or obtained from the physical hardware. Only a small number of quantities were deliberately varied during calibration. This distinction is important because a model can appear to fit the data if errors in one subsystem are hidden by changing parameters in another subsystem.

The parameters were therefore divided into three groups:

1. **Fixed hardware and baseline parameters:** These included train and wagon mass, wheel radius, gear ratio, target track voltage, motor and drivetrain parameters, and the other values already present in the inherited baseline model. They were not re-estimated during this project.
2. **Fuel-cell parameters allowed to vary:** The only fuel-cell parameters adjusted were the Tafel slope and internal resistance. These parameters control different features of the fuel-cell voltage response and were estimated by manually running the simulation with different values and observing the resulting changes against the hardware data.
3. **Migration model-form correction:** The 7N constant load was not treated as a fuel-cell parameter. It was introduced to compensate for implicit traction-load behaviour that was present in the baseline mathematical model but absent after migration to the physical Simscape network. It was selected separately and then held fixed during fuel-cell calibration.

The calibration was sequential rather than simultaneous. First, the mechanical migration was checked using an idealised power supply, and the 7N correction was established. Second, that correction and all inherited mechanical parameters were held fixed. Finally, the Tafel slope and internal resistance were manually varied to obtain the closest agreement with the measured voltage behaviour across the four train configurations.

This procedure is a practical form of sequential parameter estimation. It is not an automated optimisation routine and does not claim that the fitted values are unique physical properties of the commercial fuel-cell stack. Instead, it identifies effective parameter values that reproduce the tested system within the available operating range. This type of reduced, experimentally identified model is appropriate when detailed internal fuel-cell information is unavailable [13].

6.2 Calibrating the Fuel Cell Model

The physical Fuel Cell block in Simscape models the non-linear voltage output (V_{fc}) based on the Nernst open-circuit voltage (E_n), minus the activation losses (V_{act}), ohmic losses (V_{ohm}), and concentration losses (V_{conc}) [2], [19]:

$$V_{fc} = E_n - V_{act} - V_{ohm} - V_{conc}$$

Because the fuel cell operated primarily in the ohmic region during steady-state motion, concentration losses at extreme current limits were treated as negligible for the main calibration. The two permitted fuel-cell parameters were adjusted manually and sequentially. The final selected values were a Tafel slope parameter of $A = 0.125\text{V}$ and an internal resistance of $r_i = 0.2\Omega$.

- Tafel slope (A): This parameter governs the activation-loss curve, represented in the model by a relationship of the form $V_{act} = A \ln(i/i_0)$. Different values were tested by repeatedly running the simulation and observing how the low-current voltage transition and initial voltage response changed. The selected value was the one that produced the closest overall correspondence with the measured transition without disturbing the steady-state behavior excessively.

- Internal resistance (r_i): This parameter governs the approximately linear ohmic region, represented by $V_{ohm} = i \cdot r_i$. It was varied to control the spacing between the steady-state voltage curves of the four mass configurations. The final value reproduced the proportional decrease of approximately 0.3V to 0.5V observed when each additional unpowered wagon was added.

The calibration was based on manual parameter sweeps rather than an automated optimiser. For each trial, one parameter was changed while the other calibrated quantities were kept fixed, the simulation was rerun, and the resulting voltage curves were compared with the corresponding hardware curves. This made the effect of each parameter visible and reduced the risk of compensating for a mechanical migration error with a fuel-cell parameter.

6.3 Physical vs. Simulated Data Comparison

The four physical train configurations formed the complete comparison set for the final model assessment. The same configuration runs were used to guide the manual parameter sweeps and to assess the resulting agreement, so this should be understood as validation within the tested operating envelope rather than as an independent held-out prediction test. The purpose was to establish whether the model reproduced the measured voltage response for the locomotive alone and for the three added-mass conditions.

The transient response also presented a sampling challenge. Initial comparisons sampled the continuous simulation at exact 0.5-second intervals using a Zero-Order Hold (ZOH) to match the experimental hardware. However, this caused the simulation to miss the absolute minimum of the initial voltage dip if it occurred between sample ticks. To make the comparison more consistent, the simulation was exported continuously and a 0.5s Min-Binning technique was applied, extracting the minimum simulated voltage within each 0.5-second window.

Visual overlays of the processed experimental data against the Min-Binned simulated outputs show strong steady-state agreement while highlighting the limitations of the simplified transient model:

- Config 1: Locomotive Only

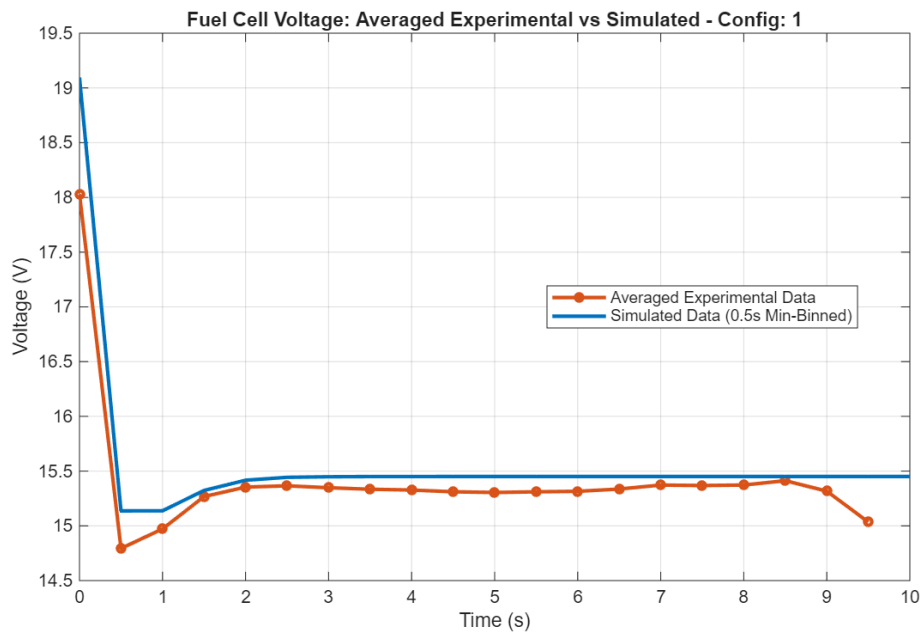


Figure 10 Actual vs Simulated Data - Locomotive Only

The lightest configuration exhibits the fastest acceleration. The simulated voltage tracks the averaged experimental data and settles into a stable steady-state level at approximately 15.5V. During the transient phase ($t=0.5s$), the experimental data dips more deeply, to approximately 14.8V, than the simulation, which reaches approximately 15.1V. This difference is consistent with the physical train having to overcome mechanical stiction in the worm-gears, a feature not represented explicitly in the simplified simulation. (Figure 10)

- Config 2: Locomotive + 1 Wagon

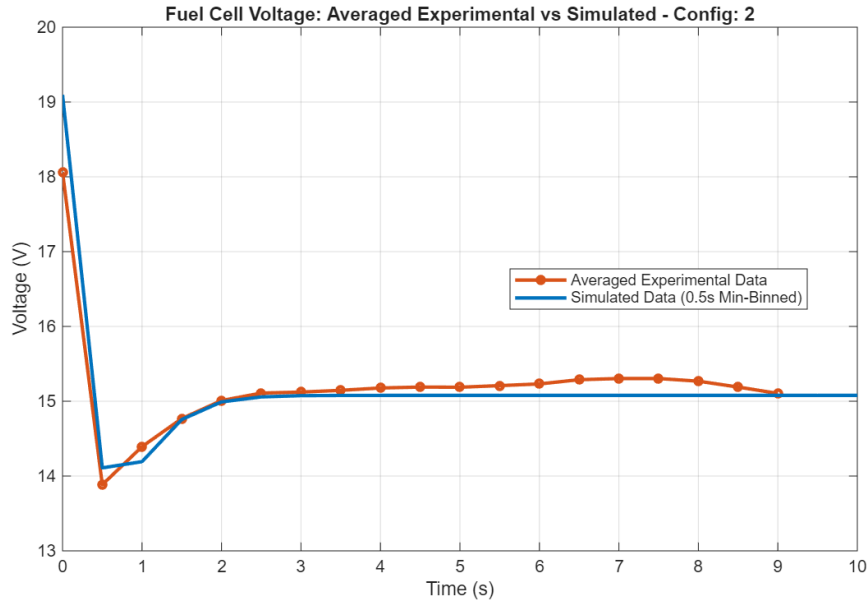


Figure 11 Actual vs Simulated Data - Locomotive & 1 Wagon

The addition of approximately 1kg of unpowered mass increases the necessary traction current. The steady-state voltage predictably drops to ~15.1V. The simulation closely reproduces this drop, supporting the effectiveness of the rigid-body mass coupling. For this medium-weight configuration, the depth of the transient dip, approximately 13.9V, is also closely matched. (Figure 11)

- Config 3: Locomotive + 2 Wagons

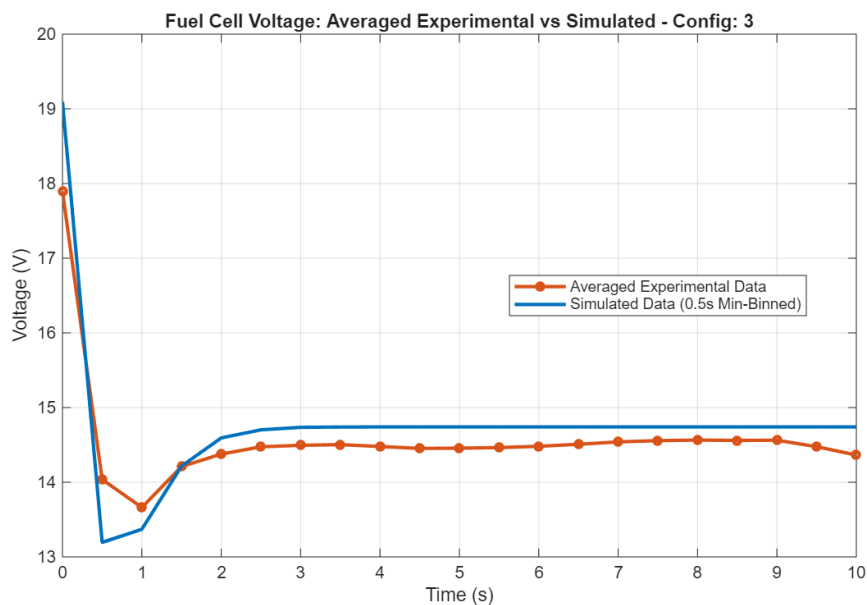


Figure 12 Actual vs Simulated Data - Locomotive & 2 Wagons

The heavy rotational inertia forces the motors to draw high current for a longer duration during start-up. The experimental transient dip reaches approximately 13.6V, while the simulation predicts a slightly deeper plunge to approximately 13.0V. Both responses recover to closely matched steady-state levels. (Figure 12)

- Config 4: Locomotive + 3 Wagons

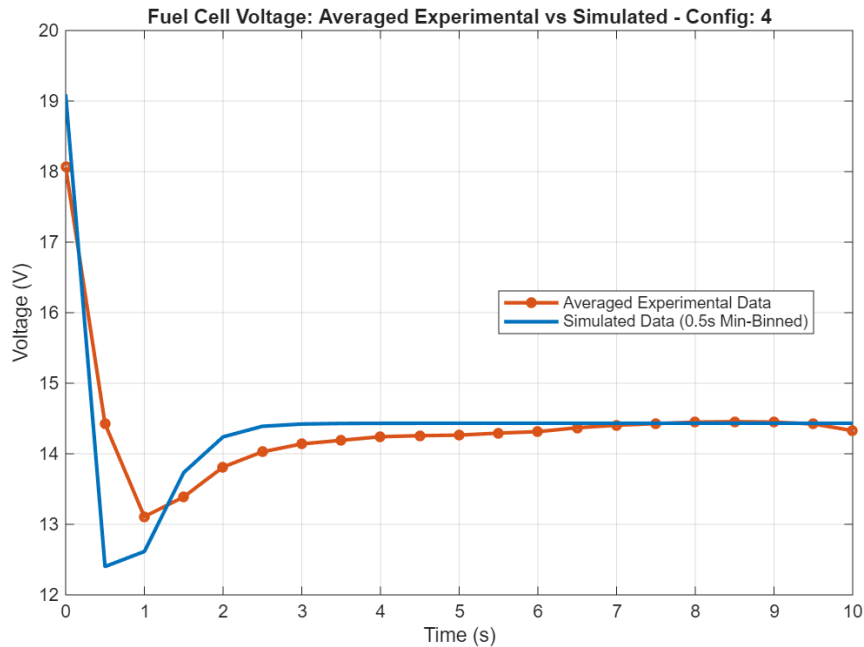


Figure 13 Actual vs Simulated Data - Locomotive & 3 Wagons

For the heaviest configuration, the train settles at approximately 14.4V (Figure 13). The transient difference is opposite to that observed in Config 1 (Figure 10): the simulation predicts an initial dip to approximately 12.4V, whereas the physical fuel-cell voltage falls to approximately 13.1V. This difference may reflect transient buffering in the real fuel-cell and converter system. Possible contributors include fuel-cell double-layer effects, converter inductance, output capacitance, and control-loop dynamics, none of which are represented in full detail by the simplified first-order model.

6.4 Statistical Validation (Transient vs. Steady-State Analysis)

To quantify the agreement between the physical model and the hardware, Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) were calculated between the simulated and experimental datasets for all four configurations. Because the calibrated parameters were selected by manually comparing these same four configuration behaviours, the results demonstrate consistency and accuracy within the tested configuration set; they do not constitute an independent generalisation test to unseen loads.

The main purpose of the project was steady-state load-dependent voltage prediction. The MATLAB Function block provides a simplified first-order voltage-reference controller for the Average-Value DC-DC Converter, and the model does not reproduce every fast electrical and mechanical transient in the hardware. A single global metric would therefore mix the primary steady-state objective with known startup limitations. Inspection of the four configuration plots showed that the main startup voltage transition had largely settled by approximately $t=2.5s$. The 10-second datasets were therefore divided into a Transient State ($t \leq 2.5s$) and a Steady-State ($t > 2.5s$). The transient metrics are reported to document the limitation, while the steady-state metrics carry the main validation significance.

Configuration	Steady-State		Transient State	
	RMSE	MAE	RMSE	MAE
Config-1 (Locomotive Only)	0.1563V	0.1315V	0.4644V	0.2951V
Config-2 (+1 Wagon)	0.1456V	0.1312V	0.4396V	0.2565V
Config 3 (+ 2 Wagons)	0.2487V	0.2430V	0.6236V	0.4639V
Config 4 (+ 3 Wagons)	0.1371V	0.1052V	0.9862V	0.7803V

Table 1 Statistical Validation of the results

6.4.1 Analysis of Steady-State Accuracy

Separating the startup and settled-motion phases allows the main validation objective to be assessed without allowing the known transient differences to dominate the results. The purpose of the steady-state assessment was to determine whether the coupled model could reproduce the change in fuel-cell voltage associated with the four different mechanical loading conditions. The results support this interpretation, although they should be understood as evidence of agreement within the tested configuration set rather than as proof of complete physical equivalence.

The steady-state MAE was 0.1315V for Configuration 1, 0.1312V for Configuration 2, 0.2430V for Configuration 3, and 0.1052V for Configuration 4. These errors are small relative to the approximately 14–15.5V operating levels observed in the experiments. The lowest steady-state MAE occurred for Configuration 4, where the error was 0.1052V on a 14.4V level, corresponding to less than 1%. Configuration 3 produced the largest steady-state MAE, which is important because it shows that the agreement is not uniformly exact across all configurations (Table 1). Nevertheless, the errors remain sufficiently small for the intended comparison of settled load-dependent voltage behaviour.

The results indicate that the model captures the total effect of increased mechanical loading on the electrical source. As wagons were added, the train required a different traction response and the measured steady-state fuel-cell voltage changed accordingly. The simulation reproduced this general load-dependent behaviour across the four configurations. This supports the effectiveness of the coupled representation of train mass, mechanical resistance, motor loading, and fuel-cell voltage response within the tested operating range.

The steady-state results do not, however, demonstrate conservation of energy or validate every individual component parameter. Only fuel-cell voltage was measured, so the experiment did not directly establish fuel-cell current, electrical power, converter losses, motor power, or total energy consumption. The agreement should therefore be interpreted as validation of the observed terminal-voltage response, not as a complete electrical or thermodynamic validation of the powertrain.

The results also provide evidence that the selected migration correction and fuel-cell parameters were effective in reproducing the measured data. They do not show that the fixed 7N correction is an independently measured physical resistance, nor that the selected Tafel slope and internal resistance are unique intrinsic properties of the commercial fuel-cell stack. These quantities were selected through manual calibration, and the same four configurations were used both to guide the calibration and to assess the final agreement. The reported metrics therefore demonstrate consistency within the tested cases rather than independent predictive performance on unseen loads.

In terms of fitness for purpose, the steady-state results support using the model as a foundation for future studies of load-dependent fuel-cell behaviour and eventual hybridisation. They do not yet establish that the model is ready to evaluate a complete Energy Management Strategy. EMS studies would require additional components and observables, including energy storage, current and power measurements, route-dependent operation, regenerative braking, and validation of transient power-sharing behaviour. The present results instead establish the source-and-load response that such future studies could build upon.

6.4.2 Analysis of Transient Limitations

The higher errors are concentrated in the transient interval defined as $t \leq 2.5$ s. The transient RMSE values were 0.4644V, 0.4396V, 0.6236V, and 0.9862V for Configurations 1 to 4, respectively. The corresponding transient MAE values were 0.2951V, 0.2565V, 0.4639V, and 0.7803V (Table 1). Because RMSE gives greater weight to larger deviations, the high Configuration 4 RMSE reflects the pronounced difference between the simulated and measured startup voltage responses. These values are consistent with the visual comparisons in Section 6.3 (Figure 13), but they must also be interpreted in light of the measurement method.

The experimental voltage was recorded at 0.5-second intervals, which limited the ability to resolve the true sub-second current and voltage behaviour at startup. The simulation was therefore processed using Min-Binning so that the minimum simulated voltage within each comparison interval could be compared with the available experimental sample. This makes the comparison more consistent with the information captured by the measurement system, but it does not recover transient information that was never recorded experimentally. The transient metrics should consequently be treated as an indication of model mismatch over the sampled startup interval, rather than as a detailed identification of the physical source dynamics.

For the lighter configurations, particularly Configuration 1, the experimental voltage dipped below the simulated voltage during startup. This pattern is consistent with the physical locomotive requiring additional breakaway effort before it begins moving. Static friction or stiction in the locomotive's worm-gear drivetrain could produce a short-lived increase in motor torque demand, which would in turn affect the fuel-cell terminal voltage. The present model does not explicitly represent this breakaway behaviour and therefore begins from a smoother mechanical response. However, this explanation remains a physically plausible interpretation rather than a demonstrated cause. No high-rate speed, motor-current, or traction-force measurement was available to isolate stiction from other possible startup effects.

The heavier configurations show a different pattern. In Configuration 4, the simulated voltage falls below the experimental voltage during the initial transient, producing the largest transient RMSE of 0.9862V. This may indicate that the physical fuel-cell and converter system provides short-timescale buffering that is not represented in full detail by the model. Possible contributors include fuel-cell double-layer behaviour, converter inductance, input or output capacitance, control-loop response, wiring resistance, and interactions between the electrical system and the mechanical load [2]. It would be too strong to attribute the discrepancy to fuel-cell capacitance alone because the experiment did not measure current and the model does not separately identify these dynamic effects.

The change in the direction of the transient error between the lighter and heavier configurations suggests that the mismatch is not caused by one simple constant voltage bias. Instead, it is likely to reflect an interaction between mechanical startup behaviour and the dynamic response of the fuel-cell powertrain. In the lighter cases, unmodelled breakaway friction may increase the initial physical load. In the heavier cases, the real electrical system may moderate the voltage response more effectively than the simplified model. The available data support these as possible explanations, but further measurements would be needed to determine their relative contributions.

The transient results therefore define an important boundary for the model's use. The model reproduces the settled voltage response well within the tested configurations, but it should not yet be presented as a detailed predictor of sub-second startup voltage, instantaneous current, electrical power, or fuel-cell energy dynamics. Higher-rate simultaneous measurements of voltage, current, motor behaviour, and train motion would be required for that purpose. Within its present scope, the model is most defensible as a steady-state load-dependent voltage model with a documented but simplified representation of transient behaviour.

7. Conclusion and Future Work

7.1 Summary of Contributions

The primary objective of this project was to construct a physically coupled simulation capable of translating the mechanical mass of a train into corresponding electrical power demand. This was achieved by migrating a functional baseline mathematical model into a hybrid physical-mathematical Simscape environment.

By replacing the original idealized voltage supply with a physical Hydrogen PEM Fuel Cell and Average-Value DC-DC Converter, the model represented a load-dependent electrical source within the mechanical train system. The 7N term was introduced as a model-form correction during migration, while the fuel-cell Tafel slope and internal resistance were manually estimated through sequential parameter sweeps with the remaining baseline parameters held fixed.

Statistical comparison using contiguous-segment alignment and 0.5s Min-Binning demonstrated strong steady-state agreement across the four tested mass configurations. With a steady-state Mean Absolute Error (MAE) below 1% under maximum load (0.1052V error on a 14.4V operating plateau), the model reproduced the measured relationship between added mechanical loading and fuel-cell voltage. The model is therefore a validated G-scale platform for steady-state load-dependent voltage analysis and a basis for future hybrid powertrain and EMS studies.

7.2 Model Limitations and Future Work

The steady-state model agreement is strong, but the project also identifies several experiments and analyses that would improve the model's physical interpretation and predictive value.

- **Mechanical stiction experiment:** The deeper measured startup dip in Configuration 1 is plausibly associated with breakaway friction in the locomotive's worm-gears. Future tests should measure repeated starts with the locomotive unloaded and loaded, while recording voltage and speed at a higher sampling rate. A stiction or breakaway-torque model could then be fitted to the measured startup response.
- **Simultaneous voltage and kinematics validation:** The present validation is based mainly on fuel-cell voltage. If simultaneously, the kinematic parameters like velocity recorded by the onboard odometry sensors are also considered, this would allow the model to be validated in a broader sense rather than from voltage alone.
- **Higher-rate transient experiment:** The 0.5-second logging interval is suitable for the steady-state objective but cannot resolve the true sub-second startup event. A faster data-acquisition system should record fuel-cell voltage, current, motor current, and track voltage simultaneously. This would help distinguish fuel-cell double-layer effects from converter inductance, output capacitance, and control-loop dynamics.
- **Independent-load validation:** The present four configurations were used to guide calibration and assess the final model. A future experiment should introduce an additional mass configuration, preferably one not used during calibration, and use it as a predictive validation case. This would test whether the model predicts correctly to an unseen load.
- **Repeatability and uncertainty experiment:** The individual runs should be analysed rather than only averaged. Repeating each configuration under the same conditions would provide standard deviations, confidence intervals, and measurement bounds for the steady-state voltage and startup dip. This would show whether model error is larger or smaller than natural hardware variation.

- **Baseline-versus-electrified model study:** The original ideal-voltage model and the final PEMFC-based model should be run under identical mechanical configurations. This comparison would quantify the benefit of replacing the infinite voltage source and show directly how the electrified model introduces realistic load-dependent voltage behaviour.
- **Route and motion extensions:** Future tests should include gradients, curves, and direct speed or odometry measurements. This would extend the model beyond the current straight, level track and allow route resistance, acceleration, and electrical loading to be validated together.
- **Hybridisation and EMS optimisation:** The validated fuel-cell model can later be extended with a battery or supercapacitor, bidirectional power conversion, regenerative braking, and an Energy Management Strategy. Such hybrid architectures and power-distribution strategies are common directions in hydrogen railway research [1], [14], [15]. The present steady-state model would provide the baseline against which transient buffering and power-split strategies could be assessed.

Overall, this project demonstrates a practical method for extending a mathematical train model into an experimentally grounded hydrogen fuel-cell traction model. Within the tested G-scale operating envelope, the model reproduced the steady-state load-dependent voltage response across four mass configurations, while the transient analysis identified the main areas requiring further refinement. Although the model is not intended to represent complete full-scale railway operation, it provides a defensible foundation for future studies of electrical power measurement, hybridisation, transient behaviour, and Energy Management Strategies.

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Appendix

- The complete directory of Simulink files, MATLAB files, images, CSV files of this project can be found in this Github repository:

<https://github.com/rahuldasi/GScaleModelTrain-Simulation.git>

- The following is the link to the Github repository of the software that was used to log the track voltage data into CSV files to be processed for validation:

<https://github.com/Niravk1997/HP-Agilent-Keysight-34401A-Control-and-Data-Logging-Software.git>